VCL2FMXConverter

Conversion Contracts Guide

Version 6.0 Navigator - detailed engineering and operator reference

Audience	Developers, maintainers, beta testers, and reviewers who need to understand or extend the v6.0 contract system.
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Project folder	C:\New Delphi Projects\VCL2FMX Converter V6
Contract root	C:\New Delphi Projects\VCL2FMX Converter V6\contracts

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1. Purpose of This Guide

This guide explains the v6.0 Navigator conversion-contract system in practical terms: what contracts are, what they contain, how the runner uses them, what they prove, what they do not prove, and how new contracts should be written when the converter learns a new rule.

The contract system exists because a source-to-source converter can look correct on one project and still fail in a different code shape. A contract captures the shape of a known problem and turns it into an executable rule. Once a problem has a contract, the converter either keeps satisfying that rule or the contract runner fails.

Plain-language definition: A conversion contract is a small source input case plus a written expectation file. The input case creates a specific conversion problem. The expectation file states exactly what the converted output and report must contain, and what must not appear.

2. What a Contract Is

A VCL2FMX conversion contract has two parts.

A fixture: one or more source files such as .pas, .dfm, .dpr, .inc, or project companion files. The fixture deliberately contains one behavior the converter must handle.

An expectation: a matching .expected.json file that records the expected conversion status, generated-code patterns, forbidden patterns, report messages, and uses-clause results.

The fixture is the test input. The expectation is the rule. Together, they become an executable contract.

Part	Example	Purpose
Fixture	contracts\colors\01_class_should_not_add_uiconsts.pas	Provides a small Pascal unit containing a class declaration and non-color words like class and declare.
Expectation	contracts\colors\01_class_should_not_add_uiconsts.expected.json	Forbids System.UIConsts and System.UITypes because the source does not use a real FMX cla color constant.
Generated output	tests\conversion_contract_output\...\output	Temporary output created by the runner so the output can be checked and discarded.
Report output	VCL_to_FMX_Conversion_Report.txt/html	The conversion report checked for required manual-review items, warnings, errors, or forbidden false positives.

What a Contract Is Not

It is not runtime data loaded by the normal converter UI.

It is not a mapping pack.

It is not a sample project meant for users to convert by hand.

It is not compared against every user project during a normal conversion.

It is not a promise that the whole converted project is finished; it proves one rule or one defined cluster of rules.

3. Why v6.0 Uses Contracts

The converter performs many different tasks: DFM parsing, FMX generation, Pascal rewriting, uses-clause cleanup, include-file analysis, Windows-message handling, reporting, mapping-pack detection, dry-run output, and more. A small change in one area can accidentally break a distant area.

Contracts reduce that risk by turning each important behavior into a repeatable check. They also make outside reviews more useful. When a reviewer finds a bug, the next question is no longer just whether the source code is wrong. The next question is whether an existing contract should have caught it.

If no contract exists, add one before trusting the fix.

If a contract exists but still passed, strengthen the contract.

If the contract fails after a change, decide whether the converter changed incorrectly or the expected behavior intentionally changed.

4. How the Contract Runner Works

The contract runner is an internal test tool. It scans the contracts folder, finds fixtures with matching expectations, runs the converter against each fixture, and compares generated files and reports against the expectation JSON.

The runner does not use a special database index. It discovers contracts directly from the folder tree. This is fast enough because each fixture is small and the directory structure is simple.

Step	What Happens	Why It Matters
1	Discover expectation files under contracts.	Each expectation identifies the input fixture to run.
2	Copy the fixture into an isolated temporary source folder.	The original contract fixture remains unchanged.
3	Run the converter through RunConversionEngine.exe.	The same core engine is exercised without clicking the UI.
4	Read generated Pascal, FMX, DPR, and reports.	The runner checks behavior, not just process completion.
5	Apply expected and forbidden regex patterns.	A contract can require correct output and forbid dangerous leftovers.
6	Print pass/fail summary.	Any unexpected behavior stops the release gate.

Normal Conversion Versus Contract Execution

A normal user conversion does not compare the user's project to every contract. The converter applies its built-in rules, mapping packs, and analysis logic directly to the user's source. The contracts are only used when the internal contract runner is executed.

In other words, if there are 19 user projects and 150 contracts, the converter does not run 19 times 150 comparisons. The contracts are only test cases for the converter itself.

5. Contract Folder Structure

The v6.0 contracts are organized by behavior family so a maintainer can find the right place for a new fixture quickly.

Folder	Purpose
colors	Color constants, UI constants, modal results, and false positives such as class and clarity.
comments_and_boundaries	Comments, strings, disabled blocks, routine boundaries, initialization/finalization boundaries, and false-positive guards.
components_and_events	Component declarations, lifecycle events, media events, image/property assignments, shape conversion, and event wiring.
dfm_fidelity	Checks for missing DFM/FMXL objects, properties, and events after generation.
dfm_pairs	Paired Pascal/DFM fixtures for FMX form generation.
graphics	Canvas/GDI conversion and visual-review reporting.
include_analysis	Pascal include directives, missing/outside/nested/recursive includes, and UTF-8 include files.
project_integration	Whole mini-project fixtures that exercise multiple systems together.
uses_clause	Unit cleanup, Windows/VCL unit preservation/removal, and conditional uses behavior.
winapi_messages	Windows messaging APIs, message constants, WndProc, message declarations, system commands, and false positives.

6. Anatomy of an Expectation File

Most expectations use the same fields. Fields can be omitted when they are not relevant to a fixture.

Field	Meaning
input_file	The fixture path relative to the project root.

expected_status	The conversion status expected by the runner, usually converted.
expected_conversions	Conversion categories expected to be reported.
expected_manual_reviews	Manual-review categories that must be present.
expected_units_added	Units that must appear in the generated uses clause.
expected_units_absent	Units that must not appear in the generated uses clause.
expected_output_patterns	Regex patterns that must appear in generated output files.
forbidden_output_patterns	Regex patterns that must not appear in generated output files.
expected_report_patterns	Regex patterns that must appear in the text or HTML report.
forbidden_report_patterns	Regex patterns that must not appear in reports.
copy_case_directory	Optional project-level mode that copies an entire fixture directory before conversion.
skip_generated_compile	Optional flag for fixtures that intentionally preserve missing or manual-review dependencies.

Example: A Simple Uses-Clause Guard

This contract proves that ordinary words beginning with cla do not inject FMX UI color units.

{

"input_file": "contracts\\colors\\01_class_should_not_add_uiconsts.pas",

"expected_status": "converted",

"expected_output_patterns": ["unit ContractClassShouldNotAddUIConsts"],

"forbidden_output_patterns": [

"System\\.UIConsts",

"System\\.UITypes"

],

"expected_report_patterns": [],

"forbidden_report_patterns": []

}

This expectation is deliberately small. It does not care about every line of generated Pascal. It cares only that the unit still converts and that two false-positive units are absent.

Example: A Positive Color Contract

A good negative contract should usually be paired with a positive contract. The negative contract says class should not trigger color units. The positive contract says a real FMX color constant should trigger them.

```
procedure TContractRealClaColorShouldAddUIConsts.Run;
var
ColorValue: TAlphaColor;
begin
ColorValue := claBlack;
Writeln(Integer(ColorValue));
end;
```

The expectation for this fixture requires both System.UIConsts and System.UITypes. Together, the positive and negative contracts prevent both over-matching and under-matching.

7. What Contracts Prove

The converter recognizes a known input shape.

The converter produces required generated-code patterns.

The converter avoids known dangerous or noisy output patterns.

The report contains required manual-review, warning, blocking, or informational messages.

The generated uses clause includes needed units and excludes unused or false-positive units.

A past bug stays fixed after future changes.

8. What Contracts Do Not Prove

They do not prove every real project will compile without manual review.

They do not replace beta testing against real applications.

They do not make unsupported VCL or Windows behavior automatically safe.

They do not prove visual correctness beyond the defined output/report patterns.

They do not prove third-party component behavior unless a mapping rule and fixture explicitly cover that behavior.

They do not run during normal user conversion unless the internal runner is invoked.

9. How Contracts Interact With Mapping Packs

Mapping packs describe third-party components and conversion actions. Contracts verify converter behavior. They are related, but not the same.

Mapping Pack	Contract
Contains JSON rules for known third-party component classes.	Contains fixtures and expectations that test converter behavior.
Used during normal conversion.	Used by the internal test runner.

Can identify, preserve, substitute, or review a component.	Can prove that mapping-pack behavior appears in generated output or reports.
Ships in the mapping_packs folder.	Ships in the contracts folder as test evidence.

When a mapping rule changes, a contract should be added or updated. The contract should prove the expected generated component, preserved property, manual-review message, or forbidden false positive.

10. How Contracts Interact With Reports

The report is part of the converter's output. For migration work, a truthful report can be as important as generated code. Contracts can therefore assert report behavior.

A real unsupported item should appear in the report.

A false positive should not appear in the report.

A blocking item should be marked as blocking when generated output cannot be trusted.

A visual-review item should still report visual review even when a safe code conversion is possible.

Informational dry-run notices should not inflate actionable issue totals.

This distinction helped v6.0 separate warnings, manual-review items, blockers, informational notices, and dry-run notices.

11. How to Add a New Contract

Add a new contract when a behavior is important enough that losing it would damage conversion quality, compile readiness, report honesty, or user trust.

Choose the correct behavior folder under contracts.

Create the smallest fixture that demonstrates the problem.

Create a matching .expected.json file.

Use expected patterns for behavior that must appear.

Use forbidden patterns for behavior that must never appear.

Run the focused contract folder first.

Run the full contract suite before trusting the change.

If the converter change touches shared logic, run regression guards too.

Example Workflow for a New Bug

Suppose a converted project reveals that TShape.Shape = stCircle becomes TEllipse correctly, but the report still asks for manual review. The proper v6.0 workflow is:

Create a fixture with a TShape whose DFM Shape property is stCircle.

Add Pascal code that assigns shpLED.Shape := stCircle.

Expect TEllipse in the generated Pascal and FMX.

Forbid Shape = stCircle in generated FMX.

Forbid FMX manual review for shpLED.Shape.

Fix the converter.

Run contracts and regression guards.

12. Good Contract Design

Good contracts are precise. They do not try to check everything. They check the behavior that matters.

Good Practice	Reason
Keep fixtures small.	Small fixtures isolate the behavior and make failures easy to understand.
Use one main rule per contract.	A contract that checks too much can fail for the wrong reason.
Include forbidden patterns.	Many converter bugs are leftovers, not missing additions.
Pair positive and negative tests.	This prevents over-matching and under-matching.
Assert report behavior.	A clean compile is not enough if the report hides migration risk.
Use project-level fixtures when needed.	Some behavior only appears when multiple units, includes, or DPR files interact.

13. Weak Contracts and How to Repair Them

A weak contract is a contract that passes even when the converter is wrong. The v6.0 color issue is a good example. The contract name said class should not add UI constants, but the expectation did not forbid System.UIConsts or System.UITypes. The runner passed because the expectation was too vague.

Rule: A contract name is not a test. Only the expectation file is a test. If the expectation does not require or forbid the right pattern, the runner cannot catch the bug.

To repair a weak contract, update the expectation so it would have failed before the fix. Then run the old behavior mentally or with the old executable if available. A repaired contract should prove that the bug is genuinely guarded.

14. Public Source Distribution Notes

The public source-distribution zip includes the contracts folder because contracts document the expected converter behavior. The public source zip intentionally excludes internal tools and generated outputs. That means the contracts are present as authoritative fixtures, but the runner scripts and RunConversionEngine.exe are not

included in that compile-only zip.

For internal development, the runner remains in the tools folder of the working repository. Before trusting contract results after engine changes, rebuild both the main converter executable and the contract runner executable.

15. Recommended Review Instruction for Outside AI Reviewers

When sending the project to another code reviewer, include the contracts folder and explain how to interpret it. A useful instruction is:

Review the converter source together with the contracts folder.

Each contract is a fixture plus an .expected.json file describing required

output, forbidden output, report expectations, and uses-clause expectations.

If you find a bug, check whether a contract already covers it.

If a contract exists but would not fail, report that the contract is weak.

If no contract exists, propose a new contract fixture and expected JSON.

16. Current v6.0 Contract Baseline

Metric	Current Value
Contract expectations	227
Mapping packs	22
Last verified full contract run	227 pass, 0 fail
Regression guard status	41 pass, 0 blockers, 0 warnings
Normal conversion dependency	Contracts are not loaded during normal user conversion.

17. Summary

The v6.0 contract system is the converter's memory. It records known migration problems as executable examples, checks generated output and reports, and prevents the same class of error from silently returning. The contracts do not remove the need for real-project testing, but they make real-project discoveries permanent. Every important converter rule should eventually have a fixture, an expectation, and a regression guard.

Maintenance Update - August 3, 2026

This maintenance update incorporates field-test feedback from real Delphi projects. The converter now recognizes additional standard VCL controls, including TBitBtn and TTrayIcon, so they are not incorrectly reported as unknown third-party components.

TBitBtn receives a conservative FMX TButton substitution note. TTrayIcon remains an explicit platform-specific manual-review item because FMX has no direct standard tray-icon component.

The current v6.0 contract suite contains 227 executable expectations. The latest verification run passed 227 of 227 contracts, and the Win32 and Win64 Release builds were rebuilt from the current source.